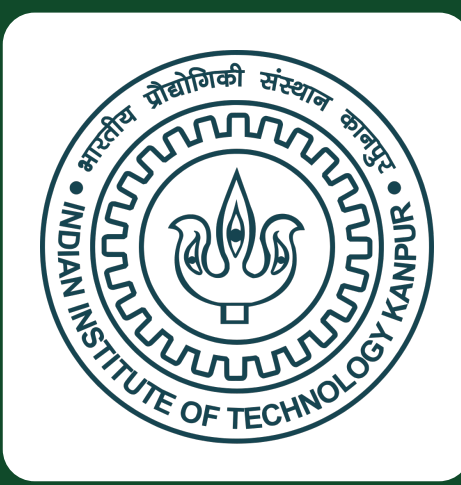


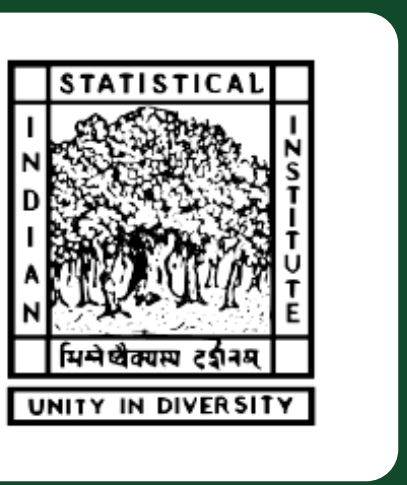


Response of Indian Iron & Steel Companies Towards Biodiversity

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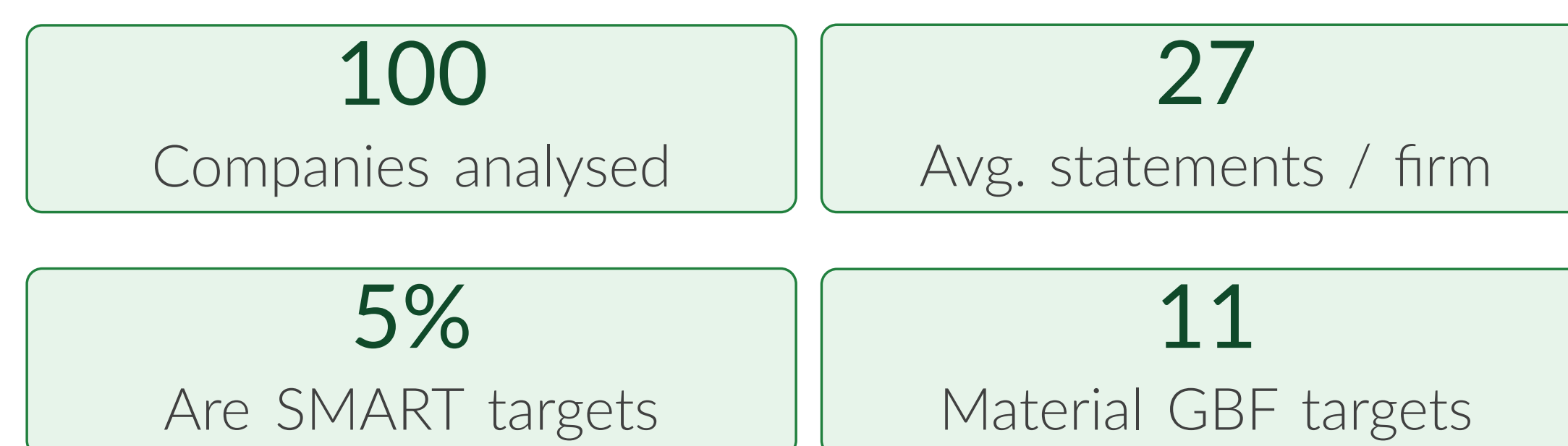
²Indian Institute of Technology Kanpur



Background & Motivation

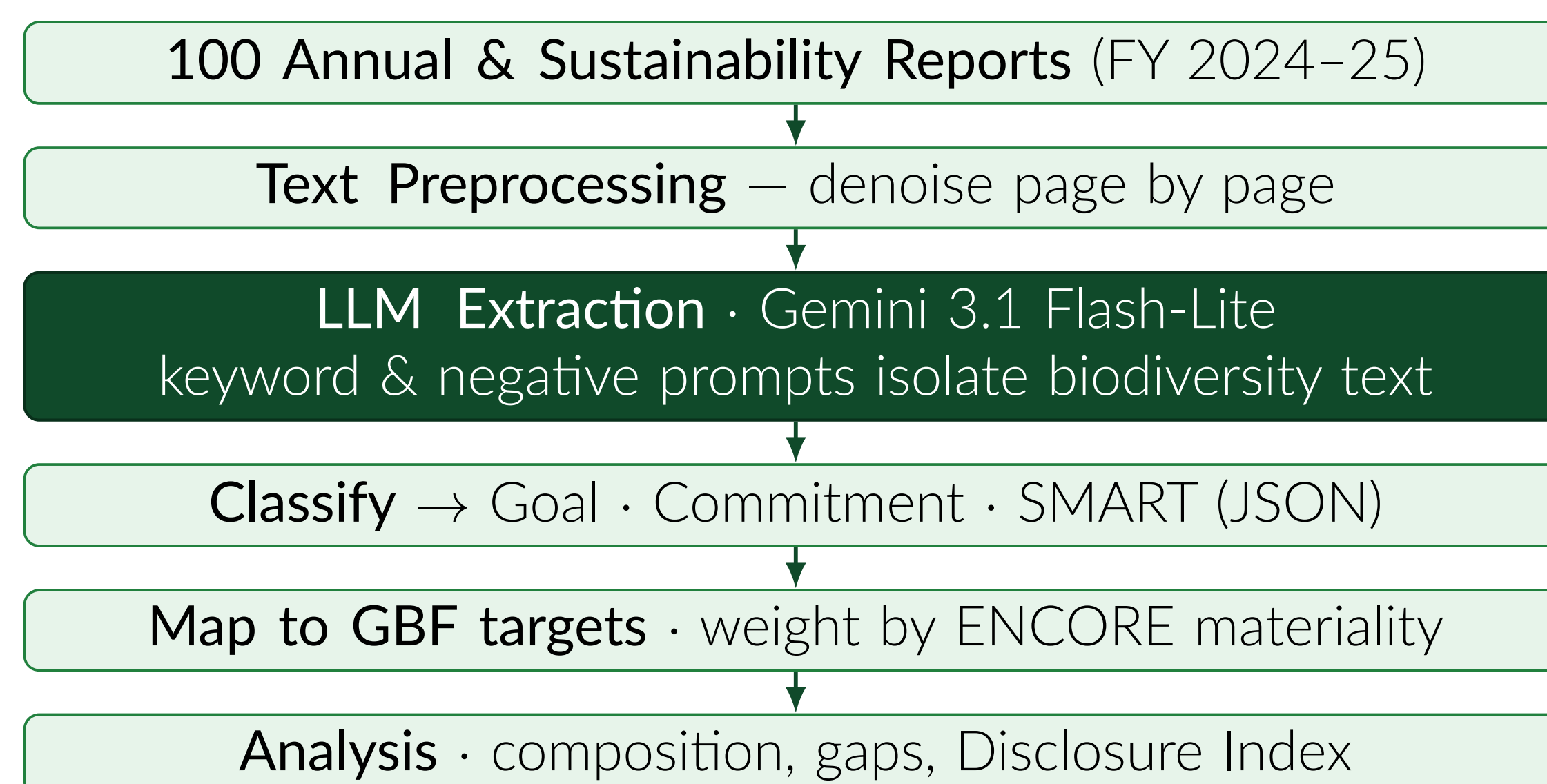
Making iron and steel depends heavily on natural resources like water supply, ecosystem services, the capacity of the environment to absorb waste, and so on. At the same time, the industry greatly harms biodiversity by extracting resources, polluting land and water, emitting GHG and non-GHG air pollutants, and degrading habitats. In 2012 the Centre for Science & Environment rated the iron sector one of the **poorest environmental performers** (“One Leaf”).

Rising stakeholder pressure now drives ESG and biodiversity disclosures. Yet it is unclear whether these address the issues *most material* to the business, or translate into **measurable action**. Manual assessments are limited and do not scale.



Methodology — an LLM Disclosure Pipeline

We built a **Large Language Model** framework that reads the FY 2024–25 reports of 100 firms (small- to large-cap; BSE / NSE / websites) and turns free text into structured, comparable data.



Frameworks & Classification

Reference frameworks

- **Kunming–Montreal GBF** — 23 global targets to halt & reverse biodiversity loss by 2030; each statement is mapped to a target.
- **ENCORE** — maps sector impacts & dependencies to a materiality $M_i \in \{1, \dots, 5\}$, used to select & weight the GBF targets relevant to iron & steel.

Every statement is classified as

- **Goal** — vague, aspirational; no metric or date.
- **Commitment** — specific intent, but rarely measurable or dated.
- **SMART Target** — Specific, Measurable, Achievable, Relevant, Time-bound.

Biodiversity Disclosure Index (BDI)

To compare firms objectively we combine the *quality* of a disclosure with the *materiality* of the issue. For each material target $i \in \mathcal{G}$ and type $* \in \{G, C, T\}$,

$$I_*(i) = \begin{cases} 1, & \text{target } i \text{ addressed by a } *, \\ 0, & \text{otherwise,} \end{cases} \quad w_G = 1, \quad w_C = 2, \quad w_T = 3.$$

With ENCORE materiality $M_i \in \{1, \dots, 5\}$, normalised by the maximum $6 \sum_i M_i$:

$$\text{BDI} = \frac{\sum_{i \in \mathcal{G}} M_i (w_G I_G(i) + w_C I_C(i) + w_T I_T(i))}{6 \sum_i M_i}$$

A high BDI rewards **measurable SMART targets on high-materiality issues**; broad goals on minor issues score low.

Disclosures Are Broad, Not Actionable

Of 100 firms, **9 disclosed nothing**; the rest ranged from 1 to 269 statements (mean 27). Reporting is dominated by aspirational language — **55% goals, 33% commitments, only 5% SMART targets**. A few large producers, led by Tata Steel (46 SMART targets of 185 statements), drive almost all measurable action.

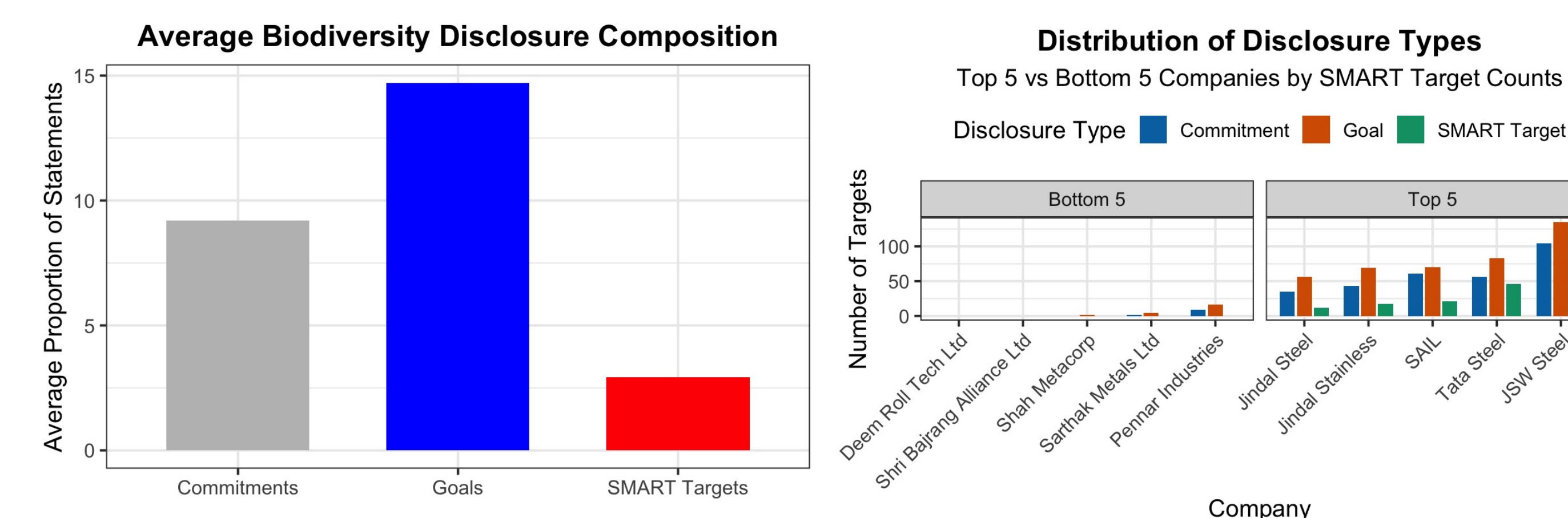


Figure 1. **Left:** average composition — goals dominate, SMART targets are rare. **Right:** top-5 vs. bottom-5 firms; even leaders keep SMART targets (green) low.

Missing the Most Material Targets

ENCORE flags **11 GBF targets** (1, 2, 3, 4, 7, 8, 11, 12, 14, 15, 16) as material, yet firms leave **~6 of 11 unaddressed** on average; only Tata Steel, Jindal Stainless, JSW Steel & APL Apollo Tubes cover all.

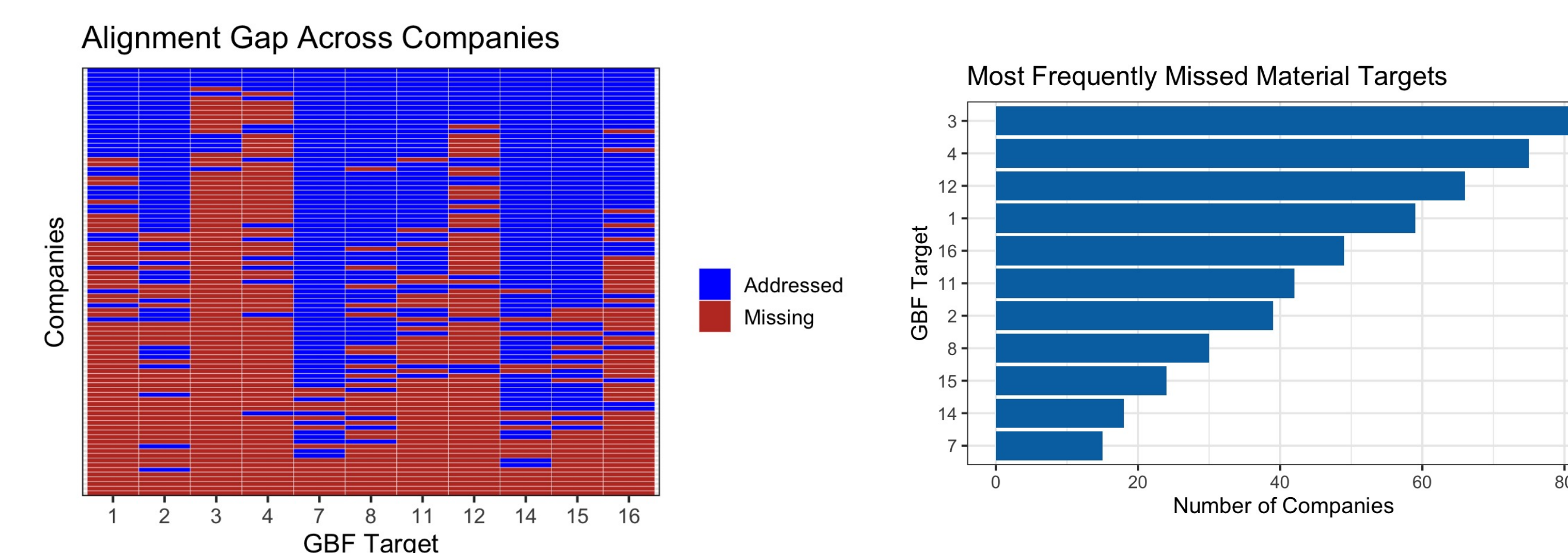


Figure 2. Company × target map (blue=addressed, red=missing). High-materiality Targets **3 & 4** are missed by ~80% of firms.

Reporting Follows Regulation

The best-covered targets — 7 (pollution) & 15 (biodiversity in governance) — are those already enforced by law: the *Environment (Protection) Act 1986*, *Air Act 1981* & *Water Act 1974* (CPCB). Targets 3 & 4 which focus on conservation of land and species diversity remain largely unattended.

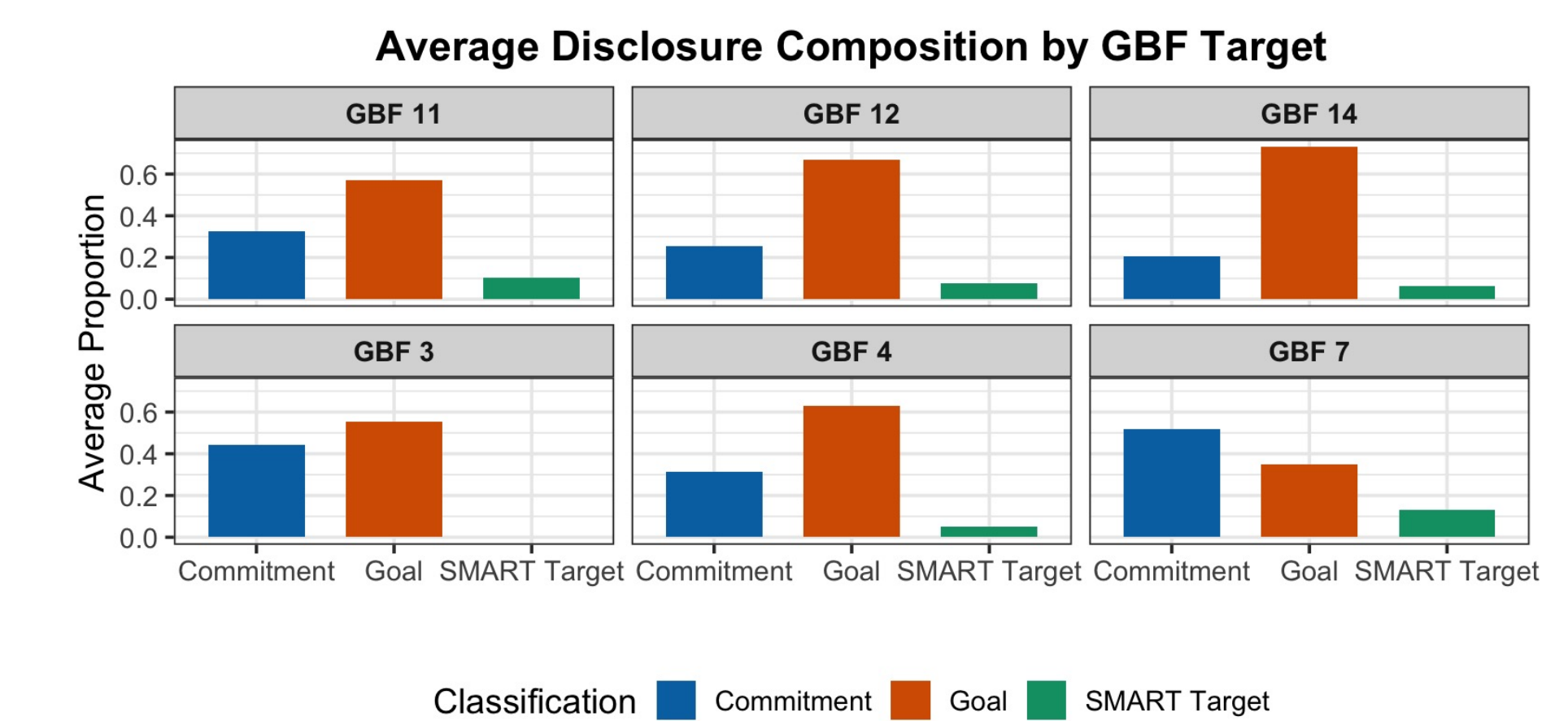


Figure 3. Composition for six high-materiality targets. Except Target 7 (pollution), **goals dominate** and SMART targets stay below 20%.

BDI Tracks Company Size & Profitability

Using Spearman’s rank correlation (robust to outliers), higher BDI is **strongly** linked to market capitalisation ($\rho = 0.85$, $p < 2.2 \times 10^{-16}$) and **moderately** to net profit margin ($\rho = 0.42$, $p < 5.6 \times 10^{-5}$). Larger and more profitable firms appear to have more biodiversity disclosures.

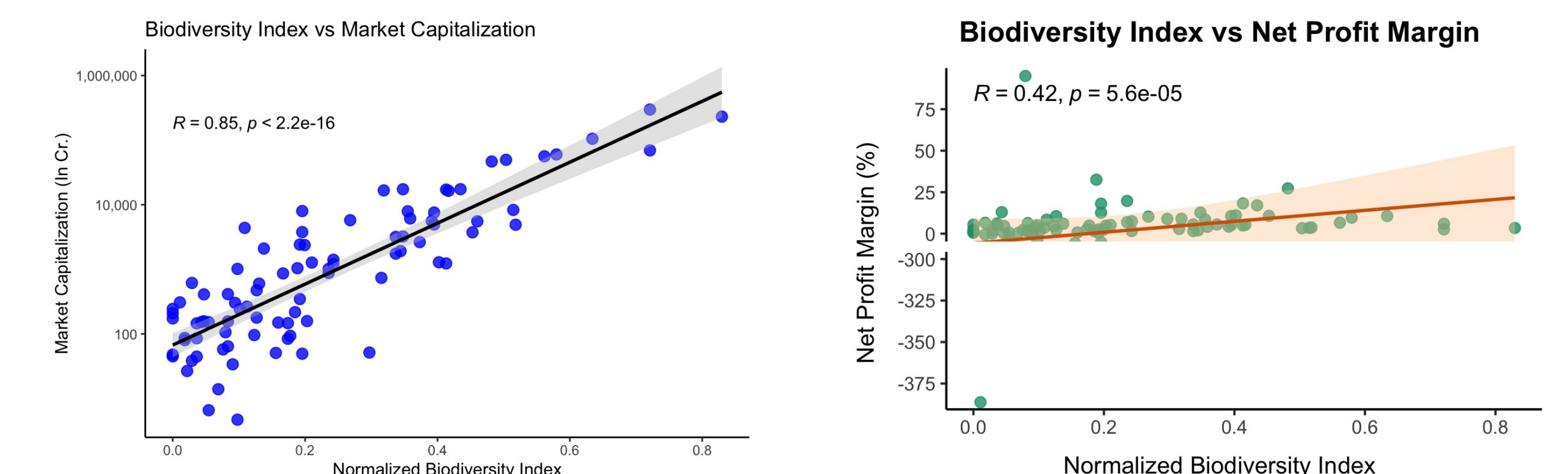


Figure 4. BDI vs. market capitalisation (left) and net profit margin (right), with fitted trends and 95% bands.

Key Findings & Policy Implications

- Firms report on issues which are already regulated. ~85% of companies talk about pollution; <20% touch genetic diversity, species extinction or land & water conservation.
- Disclosures are **broad & aspirational, not accountable** — only 5% are SMART targets.
- Regulations on species extinction and land conservation should be enforced more strictly and provide support to small firms to invest in sustainable technologies.

References

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